UNIVERSITÉ PARIS CITÉ

Faculté Sociétés et Humanités

UFR LINGUISTIQUE

Laboratoire de linguistique formelle (LLF)

Year 2026
Thesis for the diploma **Master Sciences du langage: Theoretical and Experimental Linguistics and Phonetics**
Presented and defended publicly on: **10 June 2026**

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Hybrid gender agreement in Polish

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Chapter 1

Corpus study

1.1 Methodology

1.1.1 Agreement sources

The corpus used for this study is the 300M subcorpus of the National Corpus of Polish (Przepiórkowski, 2012), which contains texts from various genres and sources, including fiction, journals, letters, Internet, spoken and educational texts, most of which were created after the year 1990. The corpus was parsed using Trankit (Nguyen, Lai, Veyseh, & Nguyen, 2021) to obtain dependency trees conforming to the Universal Dependency standard (UD). Those dependency trees were searched for words, which could trigger hybrid agreement: profession nouns, socially gendered nouns and depreciative forms of m1 nouns. The dependencies of those words then made it possible to search for attributive agreements (adjectives), predicate agreements (verbs agreeing with their subjects) and relative pronoun agreement. A tagged corpus could have been sufficient for attributive and predicate agreement, but using a parsed corpus gave a better chance of finding the correct relations and agreements within them and made it possible to include relative pronouns in the analysis. The subsections below describe how I chose the nouns for analysis.

1.1.1.1 Profession nouns

Nouns in this category came from two studies about feminitives in Polish. The first was a survey reported by Szpyra-Kozłowska (2019), where participants were asked to create feminitives from 50 masculine nouns, that in modern Polish do not have feminine counterparts at all or those counterparts are rarely used. For each noun participants had the option to give no answer. The second study was an experiment conducted by Szpyra-Kozłowska and Laidler (2022), where they tested whether certain feminitives are less accessible by asking participants to try to pronounce them. They chose 20 feminitives, which are formed from masculine nouns that have consonant clusters such as

/kt/ (e.g. *architekt*, meaning 'architect'), /tr/ (e.g. *geriatra*, meaning 'geriatrician') or /rg/ (e.g. *chirurg*, meaning 'surgeon'). According to Klemensiewicz (1957) adding the -ka suffix, the function of which can be to create the feminitve, makes the word difficult to pronounce. Those two studies give a list of 60 nouns in total, for which it is said to be difficult to form feminine counterparts. It is therefore likely that, if there was a female referent of a noun from this list, the masculine noun would be used, leading to hybrid agreement.

Both of the studies determine in some way which of the tested nouns have less available feminitives than others: in Szpyra-Kozłowska and Laidler (2022) those are the nouns for which there are the highest error rates in pronunciation, whereas in Szpyra-Kozłowska (2019) – the nouns for which the most people refused to propose a feminitive. Taking this approach has, however, two downsides: first, it requires an arbitrary threshold that separates the nouns that have an easily formed feminitive from the nouns for which creating the feminine counterpart is more difficult. Second, this approach significantly limits the number of nouns for analysis. Considering this, the current study used all of the nouns from the two studies on Polish feminitves.

Initially, there were 213 070 occurrences of agreement with the 60 nouns in this group. For ease of reference, I call all of the nouns in this category *profession* nouns, despite not all of them actually referring to professions. Meanings of all nouns are presented in Table 1.1. An example of semantic agreement with a profession noun can be seen in (1), where the masculine noun *starosta* (meaning 'mayor') has a feminine adjective *głogowska* (meaning 'of Głogów').

(1) Na zaproszenie starost-y głogowsk-iej Elżbiet-y Urbanowicz
on invitation mayor(M1)-SG.GEN of.Głogów-F.SG.GEN Elżbieta-F.GEN Urbanowicz
- Przysiężn-ej Głogów odwiedziła s. Teresa Jakubowska z
- Przysiężna-F.GEN Głogów visited sister Teresa Jakubowska from
ukraińskiego Żytomierza.
Ukrainian Żytomierz

"Invited by the mayor of Głogów, Elżbieta Urbanowicz-Przysiężna, sister Teresa Jakubowska from Żytomierz in Ukraine visited Głogów."
NCP 300M, sentence ID: 8203800

1.1.1.2 Socially gendered nouns

I chose the nouns for this category using the Polish Academy of Sciences Great Dictionary of Polish available online.[1] With the advanced search option, I searched for nouns in the thematic category HUMAN AS A PHYSICAL BEING: HUMAN PHYSICALITY: SEX. This search yielded 586 results, which then I filtered manually to find the nouns that only refer to humans (and not, for example, abstract nouns or nouns referring to body parts) and that had incongruent grammatical and semantic gender. There were 19 nouns selected this way: 7 of them were grammatically masculine animate and referred to women (e.g. *babsztyl*, meaning

[1] https://wsjp.pl/

Noun in Polish	Translation or definition in English	Noun in Polish	Translation or definition in English
adiunkt	someone who holds a PhD title and is employed at a scientific institution	architekt	architect
burmistrz	mayor	chirurg	surgeon
demiurg	demiurge	dramaturg	playwright
drań	scoundrel	dupek	asshole
dureń	fool	dziekan	dean
ekspert	expert	elekt	person elected in a specified post, but has not yet entered office
foniatra	phoniatrician	ftyzjatra	pulmonologist
geometra	geometer	geriatra	geriatrician
ginekolog	gynecologist	głupek	fool
herszt	ringleader	inżynier	engineer
jaskiniowiec	caveman	jeniec	captive
jeździec	rider	kierowca	driver
ksiądz	priest	lump	tramp
magister	master (holder of a university degree)	majster	foreman
metalurg	metallurgist	minister	minister
murarz	bricklayer	muzykolog	musicologist
mędrzec	wise man	naukowiec	scientist
nieuk	dunce	nurek	diver
pajac	clown	pastor	pastor
pediatra	pediatrician	petent	applicant
polityk	politician	porucznik	lieutenant
powstaniec	insurgent	prefekt	prefect
premier	prime minister	prezydent	president
proboszcz	parish priest	psychiatra	psychiatrist
pułkownik	colonel	rektor	person in charge of a university
sierżant	sargeant	starosta	highest administrative officer of a second-level unit of local government, mayor
subiekt	salesman	szklarz	glazier
szpieg	spy	szubrawiec	rascal
upiór	phantom	widz	audience member
wójt	highest administrative officer of a basic-level unit of local government, mayor	zbieg	fugitive

Table 1.1: Profession nouns (M1) used in the study, with their translations or definitions in English.

’woman’ in a pejorative way), 1 was grammatically masculine inanimate and referred to women (*pasztet*, referring to a woman whom the speaker finds ugly or unattractive), 7 were grammatically neuter and referred to women (e.g. *pannisko*, meaning ’a young woman’ in an expressive way), 2 were grammatically neuter and referred to men (e.g. *chłopaczysko*, meaning ’a tall or big young man’) and 2 were grammatically feminine and referred to men (e.g. *ciota*, meaning ’a homosexual man’ in a pejorative way). There were 807 occurrences of agreement with those nouns found in the corpus. The full list of nouns in this category with their translations is in Table 1.2. In (2) there is an example of feminine agreement on the word *ostatnia* (meaning ‘last’) with the masculine word *babsztyl*.

(2) Ta ostatni-a jest groźn-ym i
this(f.sg.nom) last-f.sg.nom be(prs.3sg) dangerous-m1.sg.ins and
odrażając-ym babsztyl-em
offputting-m1.sg.ins woman-m1.sg.ins
“The last one is a dangerous and offputting woman.”
NCP 300M, sentence ID: 12331643

1.1.1.3 Depreciative forms

For this category no list of nouns was created, since the corpus was annotated and I could search for the depreciative forms by checking the words’ morphological description. It is possible to create depreciative forms of surnames, however these were excluded from the analysis, because there was no way of determining the referents and therefore no way of ensuring that there is hybrid agreement. Surname *Machała* found in the corpus can serve as an example: let us say that the *Machała* family bought a new car. Assuming the family consists of at least one man, the speaker can use both (3a) and (3b) to convey that information, depending on their attitude they have towards the situation: (3a) if the speaker is neutral or positive and (3b) if they are negative. However, if there are only women in the family, the speaker can only use (3b). Without the context it is unclear whether *Machały* in (3b) is used as a depreciative or simply to refer to women and therefore we cannot decide if this is an example of hybrid agreement.

(3) a. Machał-owie kupi-li nowe auto.
Machała(m1)-pl.nom buy-pst.3pl.m1 new car

b. Machał-y kupi-ły nowe auto.
Machała-pl.nom buy-pst.3pl.nm1 new car
“Machałas bought a new car.”

Morphological analyzer Morfeusz (Kieraś & Woliński, 2017) can mark forms as surnames which made it easy to exclude them from the analysis. The other forms marked as depreciative were kept, which gave 254 different lexemes being included as depreciative forms in 1040 agreement occurrences. An example of semantic agreement with a depreciative form can be found in (4), where the noun *chłopy* (meaning ‘peasants’) in

Noun in Polish	Syntactic gender	Semantic gender	Translation or definition in English
babochłop	M2	F	a woman that resembles a man with her physicality, looks and behavior
babol	M2	F	a woman that looks or behaves in a way that causes aversion in the speaker
babon	M2	F	a woman, about whom the speaker feels negatively
babsko	N	F	a woman, about whom the speaker feels negatively
babsztyl	M2	F	a woman, about whom the speaker feels negatively
babus	M2	F	a woman, about whom the speaker feels negatively
chłopaczysko	N	M1	a tall or big young man
chłopobaba	F	M1	a man that resembles a woman with his physicality, looks and behavior (can also mean a woman that resembles a man)
ciota	F	M1	a homosexual man (can also refer to a woman about whom the speaker feels negatively)
dziewczątko	N	F	a girl, whom the speaker likes or to whom they feel compassion
dziewczę	N	F	a young girl
dziewuszysko	N	F	a girl who causes aversion or contempt in the speaker
kobiecisko	N	F	expressively about a woman, augumentative, or to refer to a big woman
kobieciątko	N	F	expressively about a woman, diminutive
kobieton	M2	F	a woman that resembles a man with her physicality, looks and behavior
pacholę	N	M1	a boy
pannisko	N	F	expressively about a young woman, augumentative
pasztet	M3	F	an ugly and unattractive woman
wamp	M2	F	a confident, misterious, provocative and attractive woman who seduces men

Table 1.2: Socially gendered nouns used in the study, with their translations or definitions in English.

depreciative form has a non-m1 ending (*chłopy* in contrast to the non-depreciative and clearly masculine *chłopi*), yet still has m1 agreement on the word *pomagać* (meaning 'help').

(4)

Chłop-y	ze	wsi	pomaga-li	tutaj.
peasants-PL.NOM.DEPR	from	countryside	helped-PST.3PL.M1	here

"Peasants from the countryside helped here."
NCP 300M, sentence ID: 9988833

1.1.2 Extracting agreement occurrences from the corpus

The code used for extracting corpus data in this study can be found in its dedicated GitHub repository.[2] The general idea of how the code works is presented in this section.

1.1.2.1 Finding the agreement controllers

After choosing the nouns to be searched in the corpus, it was necessary to find all of their forms and the forms' morphological descriptions, which I did using the morphological analyzer Morfeusz (Kieraś & Woliński, 2017). First I searched the corpus for occurrences of nouns, which could be controllers of hybrid agreement. Those nouns were saved if they were in the list of gendered or profession noun forms or if they were marked as depreciative in their morphological description. To ensure that no homonyms were accidentaly saved for later analysis, the morphological description of the found word was compared with the one given by Morfeusz. This way, for example, the word *premiera* was excluded if it was an instance of the lexeme *premiera* (meaning 'premiere') in the singular nominative form, but not if it was an instance of the lexeme *premier* (meaning 'prime minister') in the singular accusative form.

Once the right noun was found, I searched the dependency tree for words that can agree their gender with that of the noun. This was the head of the noun (which could be an example of predicate agreement), adjectives attached to the noun (which could be examples of attributive agreement), and relative pronouns referring to the noun. Based on the Universal Dependencies (UD) annotation guidelines, I searched all dependents of the word for a dependency label with a subtype :relcl, which in the UD annotation scheme marks the head of the relative clause. Then I looked for the dependent of the head of the relative clause, that had a UD feature PronType=Rel, which had to be the relative pronoun.[3]

Each occurrence was additionally automatically annotated for the distance between the controller and potential target of agreement (measured in intervening words) and the ordering: whether the target of agreement was the first or not. Once I found the possible agreement targets, I could use the data to look for occurrences of hybrid agreement.

[2] https://github.com/bmagdab/hybrid-agreement
[3] https://universaldependencies.org/workgroups/newdoc/relative_clauses.html

1.1.2.2 Extracting and listing agreement occurrences

To analyse hybrid gender agreement in the found examples, I marked the alternative to the grammatical gender of each agreement controller. For all profession nouns that alternative gender was F, because the only chance of hybrid agreement for those nouns is when the gender of the referent is a woman. For socially gendered nouns the alternative gender was assigned depending on the lexeme and for depreciative forms the alternative gender was always M1, because that is the gender of the lexemes that have depreciative forms.

Once I determined the alternative gender, I excluded plural nouns for which the gender discrepancy was between feminine and neuter. This is because for all types of agreement tested here, the plural feminine and plural neuter forms of targets of agreement are syncretic, therefore there is no hybrid agreement in those cases. This happens, for instance, in (5), where in singular ((5a) and (5b)) there is a difference between the feminine and neuter forms of the word *młoda*. In plural (5c), however, the form is the same for both neuter and feminine, so hybrid agreement cannot be observed. Examples like this were therefore excluded.

(5) a. młod-e dziewcz-ę
young-N.SG.NOM girl(N)-SG.NOM

b. młod-a dziewcz-ę
young-F.SG.NOM girl(N)-SG.NOM
"young girl"

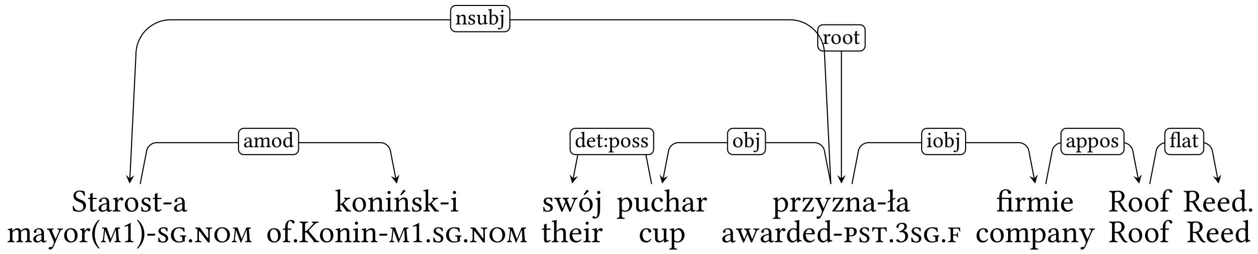
c. młod-e dziewcz-ęta
young-NM1.PL.NOM girl(N)-PL.NOM
"young girls"

The next step was to find occurrences of different types of agreement. The predicate agreement of a given noun was found in its head on three conditions: first, dependency label between the noun and the head had to be NSUBJ, which is the UD label for nominal subjects. Second, the head had to be a verb that inflects for gender, which is only true for verbs annotated as past participles,[4] active and passive participles. Third, the noun had to be in the nominative case. This condition helps filter out situations where a numeral phrase is in subject position, because then the verb does not agree with the subject – it is in 3rd person, singular and neuter form, regardless of the noun. Therefore, predicate agreement cannot be observed in those situations at all and they were excluded.

The search was simpler with attributive and relative pronoun agreement. The only requirement for attributive agreement was that the word was an adjective, which was verified by looking for the tag ADJ in the morphological description of the dependent. As for relative pronouns, all of the ones found earlier were included in the analysis, unless they did not inflect for gender (which was the case, for example, for the word *gdzie* (meaning 'where')).

[4] The annotation scheme used for NKJP uses the past participle label not only for past participles, but also for parts of past tense and conditional mood forms (Przepiórkowski, 2012).

In (1.1) we can see a sentence from the National Corpus of Polish, which serves as an example of the process of extracting agreement described in this chapter. The word, which can trigger hybrid agreement is *starosta* (meaning 'mayor'). There are two words in the sentence that agree with this word in terms of gender: *koniński* (meaning 'of Konin') and *przyznała* (meaning 'awarded'). All of the information saved for each instance of agreement is shown in Table 1.2.



"The mayor of Konin awarded their cup to the company Roof Reed."

Figure 1.1: Dependency tree for the sentence from NCP 300M, sentence ID: 10280659.

		Occurrence 1	Occurrence 2
type of text		typ_publ	
source of agreement	form	starosta	
	lexeme	starosta	
	morphological description	subst:sg:nom:m1	
	grammatical gender	m1	
	other gender	f	
agreement target	form	przyznała	koniński
	morphological description	praet:sg:f:perf	adj:sg:nom:m1:pos
	gender	f	m1
	distance	3	0
	target first	0	0
	semantic agreement	1	0
type of agreement		predicate	attributive

Figure 1.2: All of the information extracted from the sentence in (1.1)

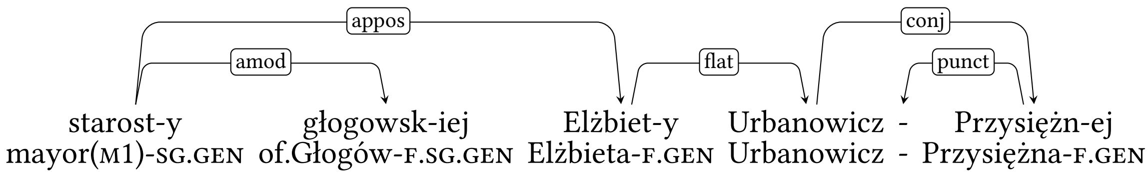
1.1.2.3 Appositions with profession nouns

Initial analysis of agreement occurrences with profession nouns showed, that most occurrences cannot be used in this study, because there is no possibility of determining whether the referent of the profession noun is male or female without the context of the surrounding text (sentences were analysed one by one). Because of this I wrote an additional script to extract information about appositions appearing with profession nouns, because if there is a feminine apposition to the profession noun, it is clear that the referent of the noun is female.[5] This is illustrated by example (1) from a previous section, repeated here. We can be sure that the M1 noun *starosta* is referring to a woman, because the noun phrase is followed by the full name of the mayor in the same case as the noun phrase.

(1) Na zaproszenie starost-y głogowsk-iej Elżbiet-y Urbanowicz
on invitation mayor(M1)-SG.GEN of.Głogów-F.SG.GEN Elżbieta-F.GEN Urbanowicz
- Przysiężn-ej Głogów odwiedziła s. Teresa Jakubowska z
- Przysiężna-F.GEN Głogów visited sister Teresa Jakubowska from
ukraińskiego Żytomierza.
Ukrainian Żytomierz

"Invited by the mayor of Głogów, Elżbieta Urbanowicz-Przysiężna, sister Teresa Jakubowska from Żytomierz in Ukraine visited Głogów."
NCP 300M, sentence ID: 8203800

The process of extracting information about agreement occurrences is almost identical to what was previously described, only here it applied exclusively to profession noun occurrences and had additional steps to find information about appositions. Using the script I first checked if the dependency head of the profession noun found in the corpus was an apposition. If that was the case, all of the dependents and the head of this apposition were treated as the dependents and the head of the profession noun and I looked for gender agreement between them and the profession noun. I calculated the distance again to make sure it is between them and the profession noun and not the apposition and I saved the word form and the morphological description of the apposition, those fields were left empty if no apposition was found. If the head of the noun was not an apposition, the noun's dependants were search to see if any of them was an apposition, in which case the procedure described above was performed. This happened in the sentence in (), and we can see the relevant part of the dependency tree in Figure (1.3).



"mayor of Głogów Elżbieta Urbanowicz-Przysiężna"

Figure 1.3: Dependency tree for a fragment of the sentence in example (1).

[5]The caveat to this approach is that it is not clear whether the feminine agreement with a profession noun is an occurrence of hybrid agreement, or if the agreement is not with the masculine profession noun but with the feminine apposition. This is further discussed in the results section.

1.1.3 Postprocessing of the agreement occurrences

The 214 917 initial observations were filtered to remove those, which were mistakenly included. There were 263 agreement observations with the gendered m3 noun *pasztet*, which as a gendered noun means a girl or a woman whom the speaker finds unattractive, but it has another, much more common meaning: pâté, a kind of forcemeat. Most of the occurrences found in the corpus were in the second meaning, not useful for the study, therefore I removed this noun completely from the study.

Some of the occurrences found had to be parsing mistakes, such as large distances between target and controller, reaching up to 1124 intervening words, relative pronouns preceding the controller and targets of agreements being comprised only of digits, which cannot have any gender agreement marking. All of these were removed.

Another group of occurrences which could not be analysed were acronyms, which were marked as depreciative forms. I removed all of them, because analyzing acronyms is outside of the scope of this study. Similar with targets or controllers of agreement being plural, which could mean that they are a part of a coordination. Coordinate structures are a separate step of Corbett's Agreement Hierarchy and also outside of the scope of this study, so I removed them as well.

Some occurrences were suspicious, because they came from a sentence which had surprisingly many agreements (up to 8). Some of them were not actual sentences, but lists of people with certain functions, for instance a list of mayors chosen in local elections in an area. I looked through sentences with unusually many agreements and removed the ones which were actually lists, since any agreement there was a parsing mistake.

In some occurrences the controller was identified by the parser to have the wrong grammatical gender. This could result in inaccurate calculations of semantic agreement, therefore I also went through them to see what should be corrected and what should be discarded. This way some of the occurrences with the noun *zbieg* (meaning 'fugitive') were removed, because it was likely that the non-m1 occurrences of this noun were used in a different meaning, for example in the collocation *zbieg okoliczności* (meaning 'coincidence'). Similarly for the word *upiór* (meaning 'phantom'), which can also be used to mean a haunting memory, for *lump* (meaning 'tramp'), which can also mean old, distressed clothing, for *pajac* (meaning 'clown'), which can also mean a kind of doll and for *nurek* (meaning 'diver'), which can also refer to the act of diving.

The last part of filtering the data concerned the gender of the target of agreement. Possible values of gendered were predefined: m1 or any other with depreciative forms, m1 or f with profession nouns and with gendered nouns it depended on the lexeme. There were some occurrences for which the target of the gender was outside of the possible scope, which was most likely a result of parsing mistakes. For depreciative nouns there was no filtering done in this part, for profession nouns I assumed everything but the m1 or f agreements was parsing mistakes and removed those occurrences, and for gendered nouns I reviewed the incorrectly marked occurrences manually.

1.1.3.1 Additional processing of profession nouns

To figure out whether an occurrence of a profession noun referred to a man or a woman, I added information about appositions attached to the nouns. I removed occurrences with plural and non-F appositions, because the plural appositions were mostly parsing mistakes and the non-F appositions show us that the referent of the noun is not a woman and there cannot be any hybrid agreement, as can be seen in ((6)). Since it is possible that in some cases we agree the target with the gender of the apposition and not the gender of the controller, I marked the ordering of the target, controller and apposition for each agreement to be able to look into the influence of the apposition. In (1) for example, the order was CTA. Only this selection of occurrences with profession nouns was saved for further analysis.

(6)

Dom	nasz	projektowa-ł	mało	znany	architekt-Ø	Stanisław
house	our	design-PST.3SG.M1	little	known	architect(M1)-SG.NOM	Stanisław
Gądzikiewicz.						
Gądzikiewicz						

"Our house was designed by a little known architect."
NCP 300M, sentence ID: 24

1.2 Results

1.2.1 Overall results

The first hypothesis this study aims to test is: there is a monotonous shift from one value of gender to another when going from attributive, through predicate to relative pronoun agreement. Additionally, the hypothesis proposed with the definition of generalized semantic agreement will be tested, which is that split hybrids (here the depreciative forms) are less likely to have generalized semantic agreement than lexical hybrids (here the socially gendered nouns and profession nouns). Before those hypotheses are addressed, we will look at the dataset extracted from the corpus.

After filtering the data which could not be used, there were 1722 observations, 443 of which were marked as semantic agreement. In Table 1.3 we can see the number of occurrences of agreement for all types of targets (attributes, predicates and relative pronouns) and the percentages of semantic agreement for all those groups.

For statistical modeling of the corpus data I used the `glmer` function from the `lme4` R package and the `glm` function from the `stats` package. I used binomial logistic regression, because the outcome was binary: whether semantic agreement was chosen or not. The first model used all of the 1722 observations. Fixed effects were the type of agreement target (attribute, predicate or relative pronoun), distance between the target and controller, target placement (preceding or following the controller), type of controller (gendered noun, profession noun or depreciative form) and apposition placement (preceding or following the controller). The corpus contains texts from 13 different types

Target of agreement	N	Semantic agreement		Syntactic agreement	
		N	%	N	%
Attribute	891	209	23.46%	682	76.54%
Predicate	686	160	23.32%	526	76.68%
Relative pronoun	145	74	51.03%	71	48.97%
Total	1722	443	25.73%	1279	74.27%

Table 1.3: Numbers and percentages of occurrences of agreement with all nouns.

of sources – this was included as the random effect in the model. The full list of coefficients in this and other models described in this chapter can be found in Appendix 2.2.2.

Types of targets and the distance were not significant as predictors of semantic agreement. Target being placed after the controller increased the likelihood of semantic agreement (p<0.001), as well as placing an apposition before the controller (p<0.001). Among the noun types the depreciative forms were the baseline, because the hypothesis was that as split hybrids semantic agreement would be less likely to occurr with them than with the lexical hybrids: gendered and profession nouns. Compared to depreciative forms then, socially gendered nouns turned out to be less likely to have semantic agreement (p<0.001), while profession noun were more likely to have semantic agreement (p<0.001).

Figure 1.4 visualises the proportions of semantic and syntactic agreement with different types of agreement, for all nouns combined and for each type of noun separately. The following subsections describe the results for each group of nouns in the study.

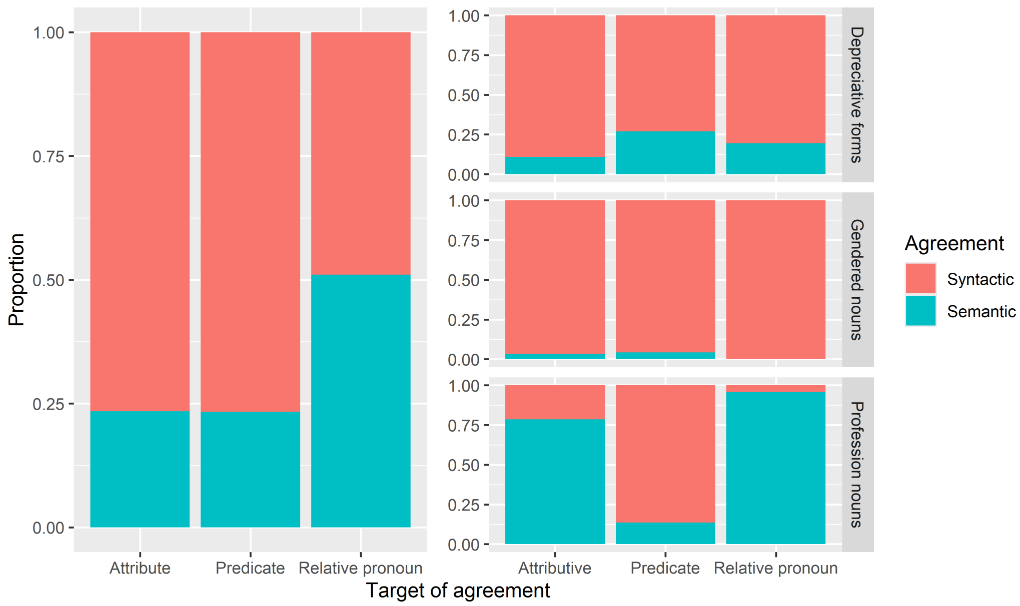


Figure 1.4: Proportions of semantic agreement for each type of target of agreement.

1.2.2 Depreciative forms

Table 1.4 shows the numbers of occurrences and percentages of semantic agreement with depreciative forms. Out of 958 occurrences of agreement with depreciative forms, 20.77% had semantic agreement. The likelihood of semantic agreement was the lowest for attributes (10.79%) and highest for predicates (26.89%), leaving in between the relative pronouns (19.57%). For this type of controller there is no monotonous change of preference for semantic agreement from attributive to relative pronoun agreement, contrary to Corbett's prediction from the Agreement Hierarchy.

Target of agreement	N	Semantic agreement		Syntactic agreement	
		N	%	N	%
Attribute	343	37	10.79%	306	89.21%
Predicate	569	153	26.89%	416	73.11%
Relative pronoun	46	9	19.57%	37	80.43%
Total	958	199	20.77%	759	79.23%

Table 1.4: Numbers and percentages of occurrences of agreement with depreciative forms.

In the following examples we can see depreciative forms with all targets of agreement in both possible forms, with the more likely form in bold. For all of them this was the form with syntactic agreement.

(7) a. **Tutej-sze**/tutej-si wykształciuch-y nie
local-NM1.PL.NOM/local-M1.PL.NOM educated.person-PL.NOM.DEPR NEG
są polskimi inteligentami.
be(PRS.3PL) Polish intellectuals

"The local educated people are not the Polish intellectuals."
NCP 300M, sentence ID: 12590744

b. Mieszczuch-y **wol-ały-by**/wol-eli-by, żeby
city.dwellers-PL.NOM.DEPR **prefer-3PL.NM1-COND**/prefer-3PL.M1-COND that
chichotał i mrugał porozumiewawczo.
giggle and wink knowingly

"The city dwellers would prefer him to giggle and wink knowingly."
NCP 300M, sentence ID: 3883941

c. Mnie najbardziej zbijają z nóg modnisi-e,
me most knock off legs dandy-PL.ACC.DEPR
któ-re/któ-rzy uważają się za orginalne.
which-NM1.PL.NOM/-M1.PL.NOM consider REFL as original

"Dandys, who consider themselves original, surprise me the most."
NCP 300M, sentence ID: 6767871

Binomial logistic regression using only the agreement occurrences with depreciative forms with the type of agreement target, distance and target placement as predictors has shown that only target placement had a significant effect on the outcome – target following the controller increased the likelihood of semantic agreement (p<0.001).

1.2.3 Socially gendered nouns

According to Corbett's Source Hierarchy, semantic agreement should be more acceptable with lexical hybrids than with split hybrids, which would mean that socially gendered nouns would have a higher percentage of semantic agreements than depreciative forms. As we can see in Table 1.5, this is not reflected in the collected corpus data, where only 3.2% of occurrences with gendered nouns have semantic agreement. Here the percentages of semantic agreement change monotonously from attributive to relative pronoun agreement, although it is the syntactic agreement that becomes more likely as we go along the Agreement Hierarchy. Considering this, the data from socially gendered nouns fits only the simple constraint of the Agreement Hierarchy (Corbett, 2023).

Target of agreement	N	Semantic agreement		Syntactic agreement	
		N	%	N	%
Attribute	343	11	3.21%	332	96.79%
Predicate	95	4	4.21%	91	95.79%
Relative pronoun	31	0	0%	31	100%
Total	469	15	3.2%	454	96.8%

Table 1.5: Numbers and percentages of occurrences of agreement with socially gendered nouns.

The following examples come from the corpus and show which of the possible agreements are preferred.

(8) a. Zachichota-ł jak
giggle-PST.3SG.M1 like
stuprocentow-a/stuprocentow-y ciot-a.
hundred.percent-F.SG.NOM/hundred.percent-M1.SG.NOM faggot(F)-SG.NOM
"He giggled like a 100% faggot."
NCP 300M, sentence ID: 18628964

b. Kobiecisk-o **przytulił-o**/przytulił-a się histerycznie do mnie.
woman(N)-SG.NOM **hug-PST.3SG.N**/hug-PST.3SG.F REFL historically to me

"The woman hugged me histerically."
NCP 300M, sentence ID: 16919204

c. Pamiętasz historię dziewczę-cia, **któr-e**/któr-a uciekło
remember story girl(N)-SG.GEN **which-N.SG.NOM**/which-F.SG.NOM ran.away
z balu?
from dance
"Do you remember the story of the girl, who ran away from the dance?"
NCP 300M, sentence ID: 18852664

Binomial logistic regression using only the agreement occurrences with socially gendered nouns with the type of agreement target, distance and target placement as predictors has shown that distance was the only significant predictor of semantic agreement: larger distance increased the likelihood of semantic agreement with gendered nouns (p<0.05).

1.2.4 Profession nouns

As we can see in Table 1.6, there were 295 occurrences of agreement with profession nouns, 77.63% of which had semantic agreement. This percentage is larger than for other types of nouns, however this is most likely because all of the profession noun occurrences included in the analysis appeared with feminine appositions, which increase the likelihood of semantic agreement. I will explore this more later when considering the order in which the apposition, controller and target appear in the sentences. With profession nouns it was more likely that semantic agreement would be used in attributes (78.54% of all attributive agreement occurrences) and in relative pronouns (95.59% of all relative pronoun agreement occurrences), but not in predicates (13.64% of all predicate agreement occurrences). This, again, does not follow the simple constraint of the Agreement Hierarchy proposed by Corbett (2023), according to which there should be a monotonic change from attributive to relative pronoun agreement.

Target of agreement	N	Semantic agreement		Syntactic agreement	
		N	%	N	%
Attribute	205	161	78.54%	44	21.46%
Predicate	22	3	13.64%	19	86.36%
Relative pronoun	68	65	95.59%	3	4.41%
Total	295	229	77.63%	66	22.37%

Table 1.6: Numbers and percentages of occurrences of agreement with profession nouns.

The preferences found in the corpus are illustrated in the following examples.

(9) a. Większość złożyli w końcówce kadencji
majority submitted in ending term
obecne-go/**obecne-j** burmistrz-Ø Stefanii Sulińskiej.
current-M1.SG.GEN/**current-F.SG.GEN** mayor(M1)-SG.NOM Stefania Sulińska
"The submitted the majority of them at the end of the term of the current mayor, Stefania Sulińska."
NCP 300M, sentence ID: 10793552

b. One są gotowe do adaptacji do roku 2008 –
they are ready for adaptation to year 2008 -
zapewniał-Ø/zapewniał-a minist-er edukacji
assured-PST.3SG.M1/assured-PST.3SG.F minister(M1)-SG.NOM education
Krystyna Łybacka.
Krystyna Łybacka
"They are ready to be adapted before the year 2008 - assured the minister of education Krystyna Łybacka."
NCP 300M, sentence ID: 4040070

c. Felicity Huffman gra panią naukow-iec,
Felicity Huffman plays lady scientist(M1)-SG.NOM
któr-y/**któr-a** w arktycznej stacji zostaje zarażona
which-M1.SG.NOM/**which-F.SG.NOM** in arctic station becomes infected
pasożytem z kosmosu.
parasite from space
"Felicity Huffman plays the scientist, who is infected with a parasite from space in the arctic station."
NCP 300M, sentence ID: 18469406

It is possible that agreement with profession nouns is more likely to be semantic than with other types of nouns, because in those occurrences there were also feminine appositions to the nouns. In those situations it is possible that speakers chose feminine agreement on targets (which for profession nouns is semantic agreement), because they were actually agreeing the gender of the target with the gender of the apposition. To have more insight into the structure of those sentences I marked the order in which the appositions (A), controllers (C) and targets (T) appeared with each occurrence of agreement. Figure 1.5 shows the proportions of semantic to syntactic agreement in occurrences with each possible order of appositions, controllers and targets.

Three of the orders shown in the figure noticeably favor semantic agreement over syntactic: apposition-controller-target, apposition-target-controller and target-apposition-controller. In contrast to the other orders, for these the apposition appears before the controller. Since this seems to be the most important information coming from the order variable, for statistical modeling instead of accounting for the order we will account for whether the apposition appears first (before the controller) or not.

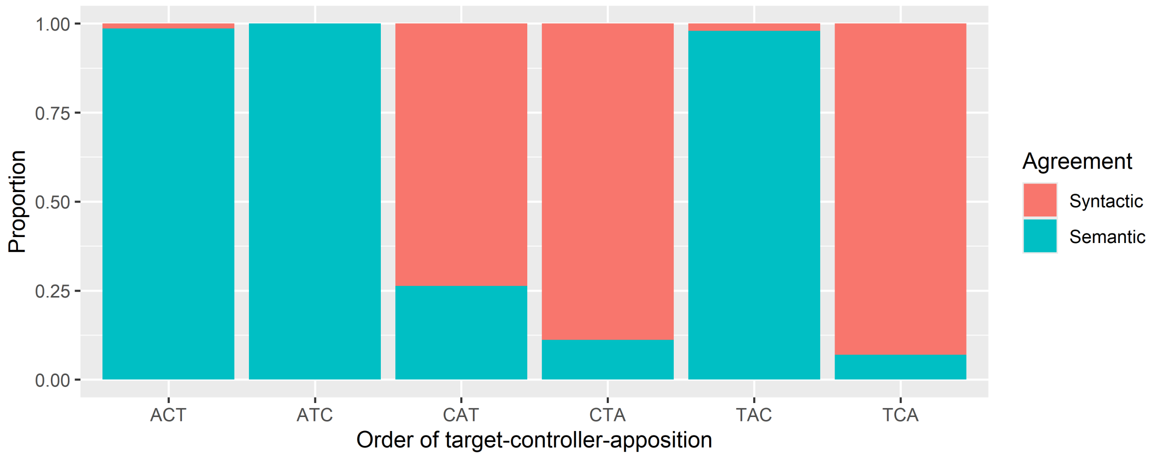


Figure 1.5: Proportion of syntactic and semantic agreements with ordering of the controller, target and apposition

The binomial logistic regression performed on agreement occurrences with profession nouns used the type of agreement target, distance, target placement and apposition placement as predictors of semantic agreement. Distance and target placement were not significant in this model. Contrary to what the Agreement Hierarchy indicates, the odds of attributive agreement being semantic were approximately 16.16 times higher than the odds of predicate agreement being semantic (p<0.05). For relative pronouns the odds were 25.84 times higher than for predicates (p<0.05). There was also a significant effect of apposition placement – the apposition appearing before the controller increased the likelihood of semantic agreement (p<0.001).

Considering the effect of apposition placement on the choice of agreement, we can look at how the proportions of semantic agreement change if we assume that we should exclude occurrences with apposition first, because the speakers agree the gender of the target with the gender of apposition and not the semantic gender of the controller. Figure 1.6 is identical to Figure 1.4, only with occurrences of profession nouns with appositions first excluded.

Target of agreement	N	Semantic agreement		Syntactic agreement	
		N	%	N	%
Attribute	46	3	6.52%	43	93.48%
Predicate	19	3	15.79%	16	84.21%
Relative pronoun	6	3	50%	3	50%
Total	71	9	12.68%	62	87.32%

Table 1.7: Numbers and percentages of occurrences of agreement with profession nouns without occurrences with appositions first.

We can see that the overall proportions of semantic agreement (left side of the figure) are now unambiguously not monotonic, and therefore not compatible with any version of the Agreement Hierarchy constraints. The proportions for the profession nouns only, however,

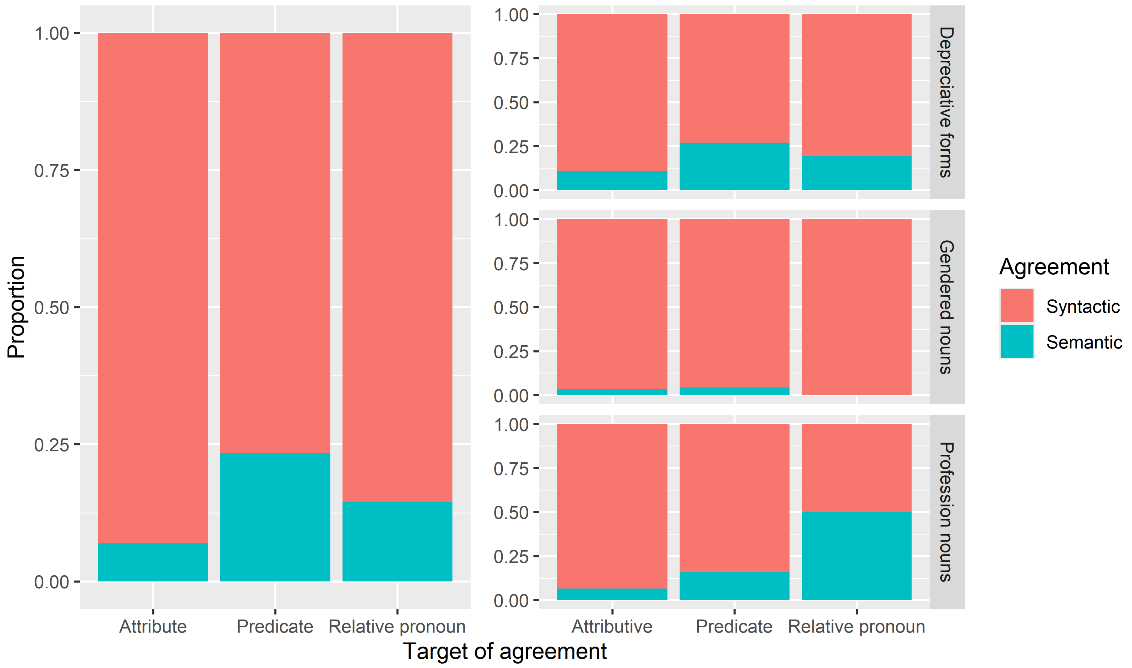


Figure 1.6: Proportions of semantic agreement for each type of target of agreement without occurrences of profession nouns with preceding appositions.

now show a monotonic shift from preference for almost exclusively syntactic agreement with attributes, to close to 50% syntactic and semantic agreement.

1.3 Discussion

When looking at all nouns combined, there was no significant effect of the type of target or the distance between the target and the controller. Instead, the choice between semantic and syntactic agreement could be better predicted with the order of target and controller, the type of noun and the apposition – whether it was there and whether it preceded the agreement controller. This is a surprising result, because the type of agreement target was hypothesised to be the main factor in the choice between semantic and syntactic agreement.

As for the effect of the type of noun, based on the Hierarchy of Agreement Sources, the lexical hybrids (profession nouns and socially gendered nouns) should be more likely to have semantic agreement than split hybrids (depreciative forms). There is a significant difference between the lexical hybrids and split hybrids, but it goes in different directions: socially gendered nouns are less likely to have semantic agreement than depreciative forms, while profession nouns are more likely to have semantic agreement than depreciative forms. Looking at this, it is inconclusive whether the lexical hybrids are more likely to have semantic agreement. However, if we exclude the profession nouns with appositions preceding the agreement controllers, we see that only 12.68% of all agreement occurrences are semantic, compared to 20.77% for depreciaitve forms. It is possible that when appositions precede the controllers, gender of targets is actually agreed with the appositions, and in this case all lexical nouns in this study have less semantic agreement than depreciative forms.

Within the depreciative forms, the only significant factor was the ordering of target and controller: target following the controller increased the likelihood of semantic agreement. This is in line with the hypothesis about the target and controller order.

Among the agreement occurrences with socially gendered nouns there is barely any semantic agreement. One reason for this could be the source of occurrences with those nouns: over 50% of those occurrences come from prose, which is highly stylized and can affect the choice of agreement with those nouns.

Profession nouns are the only group, for which the type of agreement target was a significant predictor of semantic agreement, but not in the way the Agreement Hierarchy predicts – attributive agreement was more likely to be semantic than predicate agreement. In Figure 1.6 we can see however, that if we exclude the profession noun occurrences with appositions preceding the controller, the proportion of semantic agreement changes monotonically with the smallest in attributive agreement and the largest in relative pronoun agreement. There is the possibility that with apposition preceding the controller of agreement causes speakers to choose the target gender based on the apposition and not on the controller. This seems more likely when we look at the proportions of semantic agreement in Figure 1.5 and take into account that apposition placement was a significant predictor of semantic agreement in both the general and the profession nouns model – for profession nouns only when a feminine apposition appears before the controller it is 63% more likely that there will be semantic agreement, than if the apposition was after the controller.

The main limitation here was that it was a corpus study, therefore it was impossible to control for the variables of interest. There was little data for socially gendered nouns and a lot for profession nouns, which had to be heavily filtered to be included in the analysis. Even the filtering was not fully satisfactory, since the profession noun occurrences had to have appositions included to make sure that there was in fact hybrid agreement. This in turn causes the interpretation of profession noun results to be unclear.

To continue the research on hybrid agreement in Polish, I decided to conduct two acceptability judgement experiments. I decided to use profession nouns and depreciative forms, so that it is possible to reach more definite conclusions regarding the profession nouns, and so that there is more data about split hybrids, which are a newer and less explored addition to the agreement hierarchy. The experimental study allows me to control the context, so that there is no need for using appositions and it is clear that gender agreement on the target comes from the controller.

Chapter 2

Experimental study

2.1 Methodology

We decided to test speakers' preferences for syntactic and semantic agreement in two online experiments: one using one type of lexical hybrids, profession nouns referring to women, the other using split hybrids, depreciative forms which have a shape of non-masculine nouns, but refer to men. In the experiments we tested predicative verb agreement and relative pronoun agreement.

I submitted the design of the experiments and the materials to the Paris Graduate School of Linguistics Ethics Committee and got their approval in May 2026. The following subsections describe the experiments in more detail.

2.1.1 Procedure

We conducted two acceptability judgement experiments: one using stimuli with profession nouns and the other with depreciative forms. Both experiments had the same design, which was 3x2 within-subject. The independent variables were the agreement options (semantic or syntactic) and type of agreement with ordering of target and controller: predicate with target first (prenominal), predicate with controller first (postnominal) and relative pronoun (always with controller first, since it is impossible to place the relative pronoun before its antecedent). The dependent variable was the rating of acceptability of given sentences.

The experiment was conducted in Polish. It was recommended to participants to complete the experiment using a computer or tablet, to make sure the instructions and stimuli were displayed correctly. On the first page participants were told their task was to judge whether the displayed sentences sounded naturally or not. They were also informed that participation in the study was voluntary and they could withdraw at any time. To proceed to the study they had to tick a box to confirm they consent. Next, they saw the instructions page, which had the following information.

Your task will be to judge how naturally the displayed sentences sound. If you aren't sure what this means, consider whether you could hear or say the given sentence. You will judge the sentences on a 5-point scale, from "completely unnatural" to "completely natural". You will see a control question after each sentence.

Before we proceed to the study, you will see 3 practice questions that will allow you to familiarize yourself with the task. After the practice session you will proceed to the study, which consists of 60 questions. During the study you cannot go back to the previous questions and there is no time limit to answer the questions, but answer intuitively, without thinking too much.

Out of 60 trials, 24 were with experimental items. They were arranged using the Latin square design – in 6 lists, so that each item appears on a list once and each list has 4 items per condition. Each participant was assigned a list randomly and the order of items was randomized each time. Each trial was followed by a simple comprehension question, to ensure that participants are paying attention to the experiment.

In the experiment on profession nouns the instruction in each trial was "Read the sentences below and judge how naturally the ***second*** one of them sounds", for the experiment on depreciative forms it was "Read the sentence below and judge how naturally it sounds". As was said in the instructions for participants, judgements were given on a 5-point scale. Such a scale is familiar for Polish speakers, because it is similar to the grading scale in Polish schools (1-6 with 6 being the best) and in Polish universities (2-5 with 5 being the best). The scale was labelled only on the ends with *całkowicie nienaturalne* (meaning 'completely unnatural') and *całowicie naturalne* (meaning 'completely natural').

After participants gave judgements on 60 sentences, there was a short demographic survey asking about their age, gender, first language(s) and the country or countries where they were raised. All of the response fields for those questions allowed for any kind of text answer to be typed. The duration of the whole experiment was 15-20 minutes.

2.1.2 Materials

Experimental items were constructed so that the agreement controller was either in the first or the final position in the sentence, the agreement target was immediately adjacent to the controller and there were no other words in the sentence that agreed with the controller in gender. To make sure that the last constraint was not violated, all of the sentences with relative pronoun agreement had to be in present tense (because predicates in present tense do not agree with the subject in gender), while the sentences in other conditions were in past tense. The following subsections describe how the nouns for the experimental items were selected.

2.1.2.1 Experimental items with depreciative forms

Depreciative forms appear in the paradigms of M1 nouns and morphologically they resemble nouns of genders other than M1. In Corbett's approach then, when using those nouns in a sentence we have a syntactic reason to use non-m1 agreement and a semantic reason to use m1 agreement. To make sure that semantic reason is clearly pointing to m1, the nouns selected for this experiment had to be commonly perceived as masculine. To ensure that, the selection process relied on a norming study conducted by Misersky et al. (2014), where they collected information about gender stereotypicality for role nouns in multiple Eurpean languages.

There were 22 nouns that fulfilled the following criteria:

- Perceived to be at least 60% masculine according to the norming study.
- Appear at least a 1000 times in the balanced version of the National Corpus of Polish.
- Depreciative form is not syncretic with the non-depreciative.
- Does not have disturbing, violent connotations (e.g. rapist, murderer)
- Depreciative form is not syncretic with a different noun (e.g. noun *technik* (meaning 'technician') is *techniki* in depreciative, which has the same shape as the word for 'techniques' and could lead to confusion)
- Is not a nominalized adjective (for which depreciative forms are syncretic with the feminine versions of these nouns)

Additionally, there were two nouns that did not occur in the norming study: *proboszcz* (meaning 'parish priest') and *chłopak* (meaning 'boy'). The former fulfills all of the criteria above, besides being perceived as at least 60% masculine in the norming study, since it is not included there. However, it can be included in the experiment because it is a noun commonly used in Polish and, as a subtype of a priest, it counts as being perceived as masculine.

As Skowrońska (2011) reports, the latter of the two additional nouns is problematic for Polish speakers. In a section dedicated specifically to the lexeme 'chłopak' she reports that some Polish speakers are taught that the non-depreciative form of the noun is incorrect. Because of this, hybrid agreement is very commonly used with this particular noun. The author also looked through the National Corpus of Polish to see a few examples of the noun with attributive and predicate agreement and she finds that predicates with this noun are mostly seen with semantic agreement, while attributes occur almost exclusively with syntactic agreement. This is in line with Corbett's predictions and the current study can show whether this tendency extends to the further points of the Agreement Hierarchy.

All of the nouns chosen for this experiment can be found in the appendix. In (10) there is an example of an experimental item with one of the nouns in depreciative form.

(10) Example of an item in the experiment with depreciative forms in all conditions and with its comprehension question.

a. f/m1, prenominal

Na nagrania do Ameryki Południowej poleciały/polecieli reżysery.
for shooting in America South flew:f/flew:m1 directors(m1)

The directors flew to South America for shooting.

b. f/m1, postnominal

Reżysery polleciały/polecieli na nagrania do Ameryki Południowej.
directors(M1) flew:F/flew:M1 for shooting to America South

The directors flew to South America for shooting.

c. f/m1, relative pronoun

Reżysery, które/którzy lecą na nagrania do Ameryki Południowej,
directors(M1) who:F/who:M1 are.flying for shooting to America South
nie odpowiadają teraz na maile.
NEG responding now to emails

Directors, who are flying to South America to shoot, are not responding to emails right now.

d. comprehension question
Czy w zdaniu wspomniano o Azji?
Did the sentence mention Asia?

2.1.2.2 Experimental items with profession nouns

The process of selecting nouns for this experiment was similar to that for the corpus study. The study by Szpyra-Kozłowska (2019) was used to find masculine nouns, for which it is difficult to propose a feminine version. Nouns were arranged from the most times participants of the study refused to give a feminine version of the noun, to the least. This formed a ranking, from which the following nouns were excluded:

- *ksiądz* (meaning 'priest'), *pastor* (meaning 'pastor') and *proboszcz* (meaning 'parish priest'), because in Poland the population of female priests is almost nonexistent
- offensive words (e.g. *dureń* (meaning 'fool'))
- words with less than a 1000 occurrences in the balanced version of the National Corpus of Polish (with prose excluded to make sure the words are used in everyday language)

Here each experimental item also required a sentence of context preceding the target sentence. The nouns used in this experiments are masculine and can refer to both men and women, so context is needed for participants to know that in this situation the noun refers to a woman. All of the items for this experiment can be found in the appendix and an example of an item can be found in (11).

(11) Example of an item in the experiment with profession nouns in all conditions, with its context and the comprehension question.

a. context
W tym tygodniu odwiedzają nas mamy uczniów z naszej klasy, żeby opowiedzieć o swojej pracy.
Moms of kids from our class are visiting us at school this week to talk about their careers.

b. f/m1, prenominal

We wtorek przyszła/przyszedł architekt.
on Tuesday came:F/came:M1 architect(M1)

The architect came on Tuesday.

c. f/m1, postnominal

Architekt przyszła/przyszedł we wtorek.
architect(M1) came:F/came:M1 on Tuesday

The architect came on Tuesday.

d. f/m1, relative pronoun

Architekt, która/który przyjdzie we wtorek, należy do Rady Rodziców.
architect(M1) who:F/who:M1 will.come on Tuesday belongs to council of.parents

The architect who will come on Tuesday is a member of the Parents' Council.

e. comprehension question
Czy dzień tygodnia, który był wspomniany w zdaniu, to czwartek?
Was Thursday the day of the week mentioned in the sentence?

2.1.2.3 Other items

Both experiments also had 12 control items, half correct, to establish a threshold of acceptable and unacceptable sentences. The incorrect sentences contained an attraction error with the subject, the correct ones were the same but without the error. Half of the controls had human subjects and half had non-human, this was distributed evenly within the conditions. In the profession nouns experiment these items were preceded by a sentence of context, so that the experimental item would not be too easy to distinguish. An example of a control item can be found in (12).

(12) Example of a control item in both conditions and the context sentence.

a. context
Nie wszystkie dzieci chciałyby mieć wieczne wakacje.
Not all kids want to be on holidays forever.

b. correct/incorrect

Syn sąsiadki jest szczęśliwy/szczęśliwa że idzie do szkoły.
son(M1) of.neighbor(F) is happy:M1/happy:F that goes to school

The son of the neighbor is happy to go to school.

To make sure the main focus of the study is not too obvious to participants, there were also 24 filler items. As with control items there was a context sentence preceding the fillers in the profession noun experiment. An example can be found in (13).

(13) Example of a filler item with its context.

a. context
Szukam zeszytu w którym babcia zapisywała hasła.
I'm looking for the notebook where grandma wrote down her passcodes.

b. Znalazłem na strychu sejf i nie wiem jak go otworzyć.
I.found on attic strongbox and NEG know how it open

I found a strongbox in the attic and I don't know how to open it.

2.1.3 Participants

Participants for both studies were recruited using Prolific. The target population on the platform was filtered to only include people who currently live in Poland, have Polish nationality and speak Polish as their first language. There were in total 60 participants (44 men and 16 women), their age ranged from 18 to 55. They were paid 3.20 GBP (3.7 EUR) for participation. This is 15.69 PLN for 20 minutes which corresponds to 47.07 PLN hourly, so around 150% of minimum hourly wage in Poland as of 2026.

2.1.4 Hypotheses

Based on the corpus study, one hypothesis is that syntactic agreement is preferred over semantic, so items with semantic agreement should have an overall lower acceptability rating than the items with syntactic agreement. The semantic agreement should, however, be rated higher than the ungrammatical control sentences.

Additionally, according to the stronger constraint of Corbett's Agreement Hierarchy, the acceptability of semantic agreement should increase as we go rightwards along the hierarchy. Here this means that semantic predicate agreement has a lower average acceptability rating than semantic relative pronoun agreement. Another hypothesis concerns the Hierarchy of Agreement Sources, based on which semantic agreement with split hybrids is less acceptable than with lexical hybrids. For this study the split hybrids are depreciative forms and lexical hybrids are profession nouns, therefore semantic agreement in the the profession nouns experiment should have an overall higher acceptability than semantic agreement in the depreciatives experiment. Finally, according to Corbett the target of agreement preceding the controller decreases the likelihood of semantic agreement, so in the experiments the prenominal condition should have a lower acceptability than the postnomial condition.

2.2 Results

There were 30 participants invited to participate in each study. Experiment 1 (with profession nouns) had 19 men and 11 women, Experiment 2 (with depreciative forms) had 25 men and 5 women. Participants were aged 18 to 55. Out of them, one responded that he was raised in England and Poland, three reported to speak both Polish and English as their first languages. These participants were excluded from analysis, to make sure the judgements were those of monolingual Polish speakers. All participants responded correctly to comprehension questions over 80% of the time and all of them rated the ungrammatical control questions below the grammatical ones, therefore none of them were excluded based on those criteria. There were, however, two participants in Experiment 2 that rated all experimental items as completely unacceptable. This was presumably because the depreciative forms are highly colloquial and some speakers find those forms to be ungrammatical. Those judgements were therefore not informative for the purposes of this study and responses from those two speakers were excluded. Table ? shows the final number of participants for each gender, age group and experiment.

	gender	18-30	31-50	51+
1	f	6	3	1
2	m	15	4	

	gender	18-30	31-50	51+
1	f	2	2	
2	m	15	5	1

The results of both experiments with the control conditions for comparison can be found in Figure 2.1.

We analyzed these results with a mixed-effects ordinal regression model using the clmm() function from the ordinal R package. In this model the fixed effects were the noun type (profession noun or depreciative form), agreement (semantic or syntactic), order of target and controller and the type of target (predicate or relative pronoun). The model also included interactions between the noun type and agreement and all other variables. The random effects were for participants, items, order in which items were presented, age and group from the latin square design. The full list of coefficients from this model can be found in Appendix ?.

The only significant effect in this model was the effect of noun: items with profession nouns had significantly higher ratings than items with depreciative forms ($p<0.001$). The following subsections will describe the analyses for the two experiments separately.

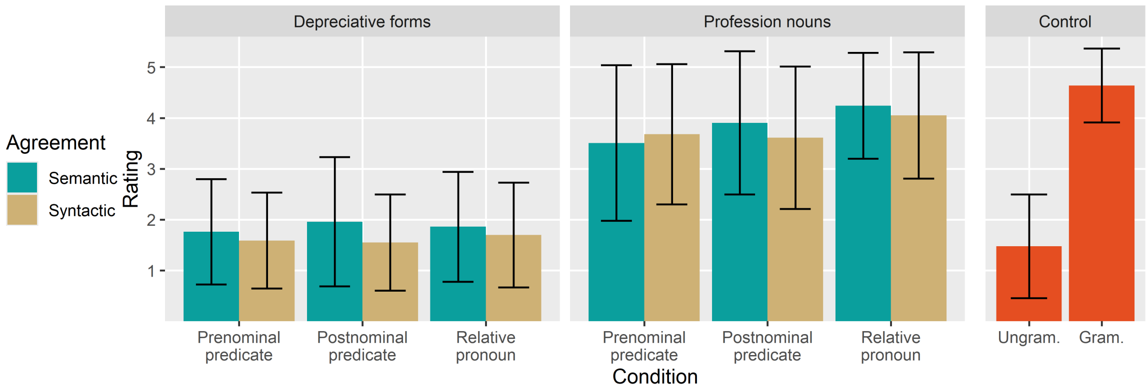


Figure 2.1: Average acceptability judgements for experimental and control conditions in both experiments.

2.2.1 Experiment 1

2.2.2 Experiment 2

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Appendices

Coefficients from statistical modeling of the corpus data

	Estimate	Std. error	Z value	Pr(>|z|)
Intercept	-5.99309	0.49265	-12.165	< 2e-16 ***
Attributive agreement	-0.36675	0.22379	-1.639	0.101
Rel. pronoun agreement	-0.34526	0.33058	-1.044	0.296
Distance	0.02303	0.03811	0.604	0.546
Controller before target	0.97644	0.21494	4.543	5.55e-06 ***
Gendered noun	-1.90220	0.28967	-6.567	5.14e-11 ***
Profession noun	1.81877	0.32474	5.601	2.14e-08 ***
Appos. before controller	4.22701	0.42209	10.015	< 2e-16 ***

Table 1: Coefficients of the mixed effects logistic regression model using occurrences with nouns of all types from the corpus study.
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

	Estimate	Std. error	Z value	Pr(>|z|)
Intercept	-5.1604	1.4187	-3.638	0.000275 ***
Attributive agreement	2.7827	1.2596	2.209	0.027158 *
Rel. pronoun agreement	3.2520	1.3174	2.469	0.013566 *
Distance	0.3212	0.2448	1.312	0.189503
Controller before target	0.9457	0.9044	1.046	0.295722
Appos. before controller	5.7027	0.7131	7.997	1.27e-15 ***

Table 2: Coefficients of the mixed effects logistic regression model using occurrences with profession nouns from the corpus study.
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

	Estimate	Std. error	Z value	Pr(>|z|)
Intercept	-4.6244	0.9319	-4.962	6.97e-07 ***
Attributive agreement	0.8582	0.8527	1.006	0.314223
Rel. pronoun agreement	-14.7725	1124.8988	-0.013	0.989522
Distance	0.4417	0.1220	3.619	0.000295 ***
Controller before target	0.2693	0.6813	0.395	0.692599

Table 3: Coefficients of the mixed effects logistic regression model using occurrences with socially gendered nouns from the corpus study.
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

	Estimate	Std. error	Z value	Pr(>|z|)
Intercept	-1.85889	0.23869	-7.788	6.82e-15 ***
Attributive agreement	-0.47954	0.24973	-1.920	0.0548 .
Rel. pron agreement	-0.61260	0.38700	-1.583	0.1134
Distance	-0.03910	0.04125	-0.948	0.3432
Controller before target	1.07946	0.24116	4.476	7.60e-06 ***

Table 4: Coefficients of the mixed effects logistic regression model using occurrences with depreciative forms from the corpus study.
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.